

# Minglai Yang

San Francisco – United States

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## Working Experience

### Scale AI

San Francisco, CA

Research Scientist, Agents

Jun 2026 – Present

- Working on agents and RL environments.

### Abaka AI

Palo Alto, CA

Senior Member of Technical Staff - Agents Lead

Jan 2026 – Jun 2026

- Led a 15-person cross-functional team on an MCP-based agent RL project, coordinating environment development, synthetic/real data generation, expert annotation, and agent evaluation workstreams.
- Designed environment simulators spanning B2B and B2C RL environments and deployed Harbor for systematic agent benchmarking and evaluation.
- Built a synthetic task and rubrics generation pipeline, producing over 3,000 high-quality, long-horizon tasks with automated evaluation criteria across 10 environments with ~800 tools in total.
- Led Dr. DocBench [2], overseeing evaluation design, data selection, and human annotation pipeline ([NeurIPS In Submission](#)).
- Co-authored research on instruction-guided video editing evaluation [3] ([COLM In Submission](#)) and LLM error verifiability [4] ([COLM In Submission](#)).

### Coretechs

Kensington, MD

Machine Learning Engineer Intern (Full Time)

May 2024 – Aug 2024

- Developed GPT-powered Slack bots with retrieval-augmented generation (RAG) and integrated real-time Slack chat into the official website using JavaScript, increasing active user engagement by 57%.
- Led end-to-end website development, from design to deployment, leveraging AWS (EC2, RDS, S3), Nginx, and Docker for scalable infrastructure.
- Conducted performance testing with Selenium, simulating 500+ users to ensure system reliability under high load.
- Received the *Best Intern Award* for outstanding contributions to the company's first GenAI project.

### The University of Arizona, AI Core

Tucson, AZ

Lead AI Engineer (Part Time)

Feb 2024 – Dec 2024

- Built tool-calling AI agents with Retrieval-Augmented Generation (RAG) for internal knowledge; trained a custom long-context documentation QA model, improving top-1 accuracy by 48% on internal benchmarks.
- Designed and shipped an end-to-end campus chatbot website serving 50,000+ UA students with long-context retrieval; increased student web engagement by 5.39× post-launch.

## Research Experience

### University of Arizona, CLU-LAB

Tucson, AZ

Undergraduate Research Assistant

Oct 2024 – Dec 2025

- Advisor: Prof. Mihai Surdeanu, Dr. Liangming Pan and Dr. Steven Bethard
- Led Research Project [1] on analyzing LLM reasoning, designed GSM-DC, a controlled benchmark for evaluating language models reasoning under irrelevant context, revealing **6 new empirical laws** on how irrelevant context disrupts reasoning, which training regimes improve robustness, and proposing test-time methods that achieve up to **6.3%** out-of-distribution performance gains ([EMNLP Oral – First Author](#)).
- Conducting research with Dr. Vikas Yadav on speculative decoding strategies for LLMs, introducing CopySpec [2], a copy-and-verify approach that delivers up to **3.08× faster generation** and **2.66× higher efficiency** in LLM self-correction reasoning ([EMNLP Oral – Second Author](#)).
- Leading a project with Prof. Steven Bethard on long-context LLM reasoning, investigating whether model reasoning is trustworthy versus prone to hallucination and other failure modes ([SemEval In Submission – First Author](#)).
- Collaborated with Dr. Chicheng Zhang to develop LLM-guided warm starts for RL [3], achieving at least 20× higher data efficiency ([EXAIT@ICML Workshop – Second Author](#)).
- Proposed using PPO to optimize the HIT@DICT reward for relation extraction; our RL optimization raised one-shot NYT29 F1 from **24.65% to 48.11%**, with more stable convergence ([ICLR In Submission – Third Author](#)).

### University of Arizona, ML4AI-LAB and IVI-LAB

Tucson, AZ

Undergraduate Research Assistant

Oct 2023 – Dec 2025

- Advisor: Prof. Kobus Barnard and Dr. Adarsh Pyarelal
- Developing dynamic Bayesian networks and Theory of Mind models within the [ToMCAT](#) project to improve AI modeling of human coordination.
- Leading a project on evaluating LLM planning capabilities under adversarial conditions, designing a controlled testbed to conduct stepwise analysis and benchmarking.
- Designing probabilistic models to disentangle causal interpersonal coordination from correlation.
- Co-authored a \$544K, two-year research proposal, focusing on LLM planning, temporal-causal reasoning, and robustness evaluation against noisy commands.

## Education

### University of Arizona

*B.S. in Computer Science*

**Tucson**

2023–2025

- GPA: 4.0 / 4.0
- Graduated December 2025, **summa cum laude** (accelerated completion of 4-year curriculum in 2 years)
- All relevant coursework available [here](#).

## Publications

1. *How Is LLM Reasoning Distracted by Irrelevant Context? An Analysis Using a Controlled Benchmark*  
**Minglai Yang**, Ethan Huang, Liang Zhang, Mihai Surdeanu, William Wang, Liangming Pan  
*EMNLP 2025 — Main Conference* [\[Oral\]](#) (CORE A\*), 2025 [\[preprint\]](#) [\[code\]](#) [\[slides\]](#) [\[poster\]](#) [\[project page\]](#)
2. *CopySpec: Accelerating LLMs with Speculative Copy-and-Paste Without Compromising Quality*  
Razvan Dumitru, **Minglai Yang**, Vikas Yadav, Mihai Surdeanu  
*EMNLP 2025 — Main Conference* [\[Oral\]](#) (CORE A\*), 2025 [\[preprint\]](#) [\[code\]](#)
3. *Improving the Data-efficiency of Reinforcement Learning by Warm-starting with LLM*  
Thang Duong, **Minglai Yang**, Chicheng Zhang  
*Exploration in AI Today (EXAIT) Workshop @ ICML, 2025* [\[preprint\]](#) [\[code\]](#)
4. *AlignSAE: Concept-Aligned Sparse Autoencoders*  
**Minglai Yang**, Xinyu Guo, Zhengliang Shi, Jinhe Bi, Steven Bethard, Mihai Surdeanu, Liangming Pan  
*Transactions on Machine Learning Research (TMLR)*, 2026 [\[openreview\]](#) [\[preprint\]](#) [\[code\]](#) [\[project page\]](#)
5. *EchoRL: Reinforcement Learning via Rollout Echoing*  
Jinhe Bi, Aniri, **Minglai Yang**, Xingcheng Zhou, Wenke Huang, Sikuan Yan, Yujun Wang, Zixuan Cao, Michael Färber, Xun Xiao, Volker Tresp, Yunpu Ma  
*International Conference on Machine Learning (ICML)*, 2026 [\[preprint\]](#)

## Preprints

1. *Peeking inside the Black-Box: Reinforcement Learning for Explainable and Accurate Relation Extraction*  
Xinyu Guo, Zhengliang Shi, **Minglai Yang**, Mahdi Rahimi, Mihai Surdeanu  
*ICLR In Submission*, 2026 [\[preprint\]](#)[\[code\]](#)
2. *Dr. DocBench: A Comprehensive Benchmark for Expert-Level and Difficult Document Parsing*  
**Minglai Yang**, Xinyan Velocity Yu, Pengyuan Li, Xinyu Guo, Zhenting Qi, Konwoo Kim, Longtian Ye, Xiaolong Luo, Jinhe Bi, Henry Zhang, et al.  
*NeurIPS In Submission*, 2026 [\[preprint\]](#)
3. *VEFX-Bench: A Holistic Benchmark for Generic Video Editing and Visual Effects*  
Xiangbo Gao, Sicong Jiang, Bangya Liu, Xinghao Chen, **Minglai Yang**, Siyuan Yang, Mingyang Wu, Jiongze Yu, Qi Zheng, Haozhi Wang, Jiayi Zhang, Jie Yang, Zihan Wang, Qing Yin, Zhengzhong Tu  
*COLM In Submission*, 2026 [\[preprint\]](#)
4. *Justified or Just Convincing? Error Verifiability as a Dimension of LLM Quality*  
Xiaoyuan Zhu, Kimberly Le Truong, Riccardo Fogliato, Gokul Swamy, Weijian Zhang, **Minglai Yang**, Longtian Ye, Bangya Liu, Minghao Liu, Andrew Ilyas, Steven Wu  
*COLM In Submission*, 2026 [\[preprint\]](#)
5. *Opening the Black Box: A Survey on the Mechanisms of Multi-Step Reasoning in Large Language Models*  
Liangming Pan, Jason Liang, Jiaran Ye, **Minglai Yang**, Xinyuan Lu, Fengbin Zhu  
*arXiv preprint*, 2026 [\[preprint\]](#)

6. *Knowledge Index of Noah's Ark*  
Sheng Jin, Minghao Liu, Yunze Xiao, et al. (incl. Minglai Yang)  
*arXiv preprint*, 2026 [\[preprint\]](#)
7. *Triaging Threats to Specialized Guardrails*  
Wenjie Jacky Mo, Xiaofei Wen, Rui Cai, Boyu Zhu, Sicong Jiang, Zihan Wang, Minglai Yang, Zhe Zhao, Muhao Chen  
*arXiv preprint*, 2026 [\[preprint\]](#)
8. *Word2VecGD: Neural Graph Drawing with Cosine-Stress Optimization*  
Minglai Yang, Reyan Ahmed  
*arXiv preprint*, 2025 [\[preprint\]](#)

## Honors and Awards

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**2025: Best Senior Award**, *University of Arizona*

**2025: 2nd Place — Reddit Wildcat Hackathon** *Reddit & University of Arizona*

**2025: Undergraduate Research Travel Grant (Top 10 awardees university-wide)**, *University of Arizona*

**2025: Galileo Circle Scholar (Top 0.8% Academic Award)**, College of Science, *University of Arizona*

**2023–2025: Academic Highest Distinction (Dean's List) (6 times)**, *University of Arizona*

**2023–2025: Global Wildcat Scholarship**, *University of Arizona*

**2021: Shanghai High School Science and Technology Competition**, Second Prize, *Shanghai High School*

**2020: Shanghai Youth Science and Technology Innovation Competition**, Third Prize, *Shanghai, China*

## Leadership and Extracurricular Activities

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### AI Club, University of Arizona

**Tucson, AZ**

*President*

*Aug 2024 – Dec 2025*

- Advisor: Dr. Zhuolin Yang, Dr. Adarsh Pyarelal and Prof. Kobus Barnard
- Organized workshops on AI agents and LLM applications, including a Hack Arizona event with 100+ participants.
- Led weekly lectures and invited speaker sessions, engaging over 40 students across disciplines to foster AI learning.
- Organized a reading group that mentors students interested in AI research.
- Raised over \$14,000 in sponsorships to support club activities and student research projects.

### Mathematical Contest in Modeling (MCM), University of Arizona

**Tucson, AZ**

*Team Leader & First Author*

*Sep 2023 – Mar 2024*

- Led weekly tutoring sessions on mathematical modeling, convex optimization, Bayesian inference, and deep neural network methods.
- Developed mathematical models for the 2024 MCM problem on sea lamprey ecosystems, focusing on resource availability and sex ratio dynamics.

## Teaching

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**Fall 2025:** TA, CSC-244: Discrete Mathematics II, Computer Science, University of Arizona

**Fall 2025:** Instructor, Deep Learning and Natural Language Processing, AI Club, University of Arizona

**Spring 2025:** TA, CSC-144: Discrete Mathematics I, Computer Science, University of Arizona

**Fall 2024 – Spring 2025:** Instructor, Math for AI Workshop Series, AI Club, University of Arizona

## Service

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*ACL Reviewer*

*2026*

*ICML Reviewer*

*2026*

*AI4MATH Workshop @ ICML Reviewer*

*2024*

## Technical Skills

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**Languages & Systems:** Python, Java, C/C++, JavaScript, MATLAB, Ruby, Swift; Linux, Git, Docker, AWS, Nginx

**ML/DL & Finetuning:** PyTorch, NumPy, pandas, scikit-learn, PyMC; PEFT (LoRA/QLoRA, Prefix/P-Tuning), SFT/DPO/PPO, RLHF/RLAIF; LLaMA-Factory, VERL, OpenRLHF, DeepSpeed-Chat; bitsandbytes

**NLP & RAG/Serving:** Hugging Face (Transformers/Datasets/Tokenizers), spaCy, NLTK, SentencePiece, SentenceTransformers; LangChain, LlamaIndex, FAISS, Milvus, Qdrant, Weaviate, Chroma, pgvector; Elasticsearch/BM25, vLLM, FlashAttention, TransformerLens